

ABSTRACT

Earth observation (EO) satellite data has become a powerful resource for monitoring land cover, environmental change, and urban growth at scale. Yet its practical use is often limited by complex preprocessing requirements, the heavy computational demands of modern machine learning models, and the lack of accessible tools for domain experts who are not software specialists.

This thesis presents the design and implementation of an end-to-end EO analysis pipeline that bridges the gap between raw satellite imagery and actionable geospatial intelligence. The system integrates harmonized Landsat and Sentinel-2 (HLS) data with the NASA-IBM Prithvi-300M foundation model - one of the largest geospatial machine learning models trained specifically on EO imagery. The pipeline automates ingestion of multi-source satellite data, performs spectral band harmonization, and executes GPU-accelerated land cover segmentation across four primary classes: vegetation, urban areas, water bodies, and barren land.

Processed outputs are converted into Cloud-Optimized GeoTIFFs and published as Web Map Service layers via GeoServer, enabling seamless integration with standard GIS workflows. A web-based dashboard built with Streamlit provides an intuitive interface for non-technical users to upload imagery, track processing status, and interactively visualize classification results on geographic maps.

Experimental results confirm accurate multi-class segmentation across diverse landscapes. The system demonstrates that foundation models can be effectively deployed in production EO pipelines, reducing both the time and expertise required for large-scale land cover analysis. These findings highlight the potential of AI-driven EO systems to support environmental monitoring, urban planning, and disaster response applications.